

The Power Of GRID COMPUTING



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The concept of tapping unused CPU cycles was born in the early 1970s. This was the time when computers were first getting linked together using network engineering. In 1973, Xerox's Palo Alto Research Center (PARC) installed the first Ethernet network, and the first full-fledged distributed computing effort was underway. Scientists John F. Shoch and Jon A. Hupp developed a worm (as they named it) and imagined it moving from machine to machine using idle resources for doing something resourceful. Richard Crandall, an eminent scientist at Apple, started putting idle, networked NeXT computers to work. He installed software that enabled the machines to perform computations and to combine efforts with other machines on the network, when not in use.

The term grid computing came up much later in early 1990s as a metaphor for making computer power as easy to access as an electric power grid. CPU scavenging and volunteer computing became quite popular in 1997 courtesy *distributed.net*. Two years later, SETI@home project was established to tap the power of networked PCs worldwide, with the objective of solving CPU-intensive research problems.

Ian Foster, Carl Kesselman and Steve Tuecke are widely regarded as the fathers of the grid, as they brought to-

gether the ideas of the grid (including those from distributed computing, object-oriented programming and Web services).

They pioneered the creation of Globus toolkit that incorporates not only computation management

but also storage management, security provisioning, data movement, monitoring and a toolkit. This was used for developing additional services based on the same infrastructure, including agreement negotiation, notification mechanisms, trigger services and information aggregation.

Globus Toolkit remains the *de facto* standard for building grid solutions. In 2007, the term Cloud computing became a buzz word and is conceptually similar to the canonical foster definition of grid computing.

What grid computing is all about

Grid computing is a type of distributed computing that comprises coordinating and sharing computing, application, data, storage or network resources across dynamic and geographically spread out firms. Basically, it is a computer network in which each computer's resources are shared with every other computer in the system, that is, processing power, memory and data storage are all unrestricted resources that authorised users can access and control for certain projects.

A grid computing system can be elementary and homogenous such as a pool of similar computers running on the same operating system (OS), or complex and heterogeneous such as inter-networked systems consisting of nearly every computer platform that exists.

Grid computing started as a response to scientific users' need to combine huge amounts of computing power to run very complex applications. The *ad hoc* assemblages of distributed resources were coordinated by software that mediated the various OSes and managed aspects like scheduling and security to create sophisticated, virtual computers.

Grid computing remains confined more to the research community and is a sign of utility-style data processing ser-

Ian Foster, Carl Kesselman and Steve Tuecke are regarded as the fathers of the grid (Image courtesy: <http://www.calit2.net>)

